

A surface flow line model of a scratching tip: apparent and true local friction coefficients

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Abstract

The apparent friction coefficient is the ratio between the tangential force and the normal load applied to a moving scratching tip. It includes a so-called “true local” friction coefficient, which is the scission at the interface between the tip and the surface being scratched, and a “geometrical” friction coefficient, which is the plough effect due to the wave front created ahead of the moving tip and depends on the shape of the tip. Like in any mechanical test, three basic types of behaviour of the material at the interface are observed: purely elastic, elastic–plastic and fully plastic. As is usual in polymers, the material behaviour is time and temperature dependent and may exhibit strain hardening. A surface flow line model is developed here to deduce the geometrical and true friction coefficients at the interface between a moving scratching tip and a surface from the apparent friction coefficient. Using this model, several situations may be simulated to predict the influence of the geometry of the tip on the scratch resistance of the material.

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1. Introduction

1.1. Scale effects in friction mechanisms

Historically, there were two approaches to friction. The first was based on statistical mechanical models of the infinite number of microscopic contact points between two surfaces and the second on the concept of adhesion between surfaces. The link between these two approaches was first established by Bowden and Tabor [1] by taking into account the mechanical contact and adhesion between microscopically rough surfaces. Unfortunately, none of these models is able to predict quantitatively real friction coefficients. The macroscopic friction coefficient is the sum of the interactions between a large numbers of microscopic elementary contacts, but the friction of each of these elementary contacts is only an apparent value. This apparent value is itself the sum of a true local friction (the ratio

between the interfacial shear stress at the interface and the local pressure) and the viscous and/or elastic–plastic and/or plastic flow of the material around the local contact area. Consequently, there are three main scales in the overall friction mechanism: the macroscopic scale, the apparent local scale and the interfacial scission true friction.

The most complicated situation arises at the local interfacial level where either mechanics or thermodynamics control the friction. The earliest models of the macroscopic contact [2] were developed for static contacts and perfectly elastic–plastic materials and neglected any adhesion phenomena. These models are consistent with the Bowden and Tabor analysis connecting normal load and contact area and also predict the loading level up to which the contact remains perfectly elastic. They have been refined on the local scale by accounting for adhesion [3,4] and introduced into macroscopic statistical models describing rough surfaces [5,6]. Such a statistical model was recently used to describe polymer–polymer friction [7].

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Nomenclature

μ_{app}	apparent friction coefficient
μ_{ad}	adhesion friction coefficient
μ_{def}	strain friction coefficient (ploughing friction coefficient in the case of scratching)
μ	true friction coefficient
a_r	length of the rear contact
a_f	length of the front contact
a	contact radius (case of spherical tip) or frontal contact length (case of wedge)
b	rear contact length (case of wedge)
$\chi = b/a$	characterises the symmetry of the contact area (case of wedge)
S_d	rear contact area
S_f	frontal contact area
l	half width of the groove behind the tip
ω	angle describing the symmetry of the contact area
Y	yield stress
E^*	normalised elastic modulus at the contact ($\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$)
c	ratio of the yield stress measured under compression to the scratch hardness ($c \approx 2$ for polymer)
ε	contact strain
ε_e	highest possible strain of the elastic sliding contact
ε_p	lowest possible strain of the plastic contact
F_n	normal force acting on the tip
F_t	tangential force acting on the tip
V_{tip}	sliding speed of the moving tip
$\vec{x}, \vec{y}, \vec{z}$	vectors of the moving tip
$\vec{z}$	from the tip towards the surface
$\vec{x}$	direction of movement of the tip
$\vec{N}$	elementary normal load vector
$\vec{T}$	elementary tangent load vector
$\vec{n}$	normal unit vector
$\vec{t}$	unit vector tangential to the flow lines
τ	interfacial shear stress
p	local normal pressure stress
φ	slope of the contact pressure distribution
C	centre of the sphere at the end of the tip
O	centre of the circle connecting the spherical end and the cone
r	radius
R	radius of the tip
θ	half angle of the cone or half angle of the wedge
$R_c = R \cos \theta$	radius of the circle connecting the spherical end and the cone
A, B, C, D	local pressure and shear elementary action integrals in the z and x directions
$\alpha, \beta, \gamma, \alpha_t$	angular variables of the components of the local pressure and shear stresses
$\alpha_r, \alpha_l, \gamma_l$	angular limits of the contact area
δ	angle

However, to explore what happens at the interface (interfacial shear stress), it is much more convenient to consider perfectly smooth surfaces with one single contact, an approach which does not require statistical models. The adhesion was recently introduced into the interface in local models of a scratching process [8,9]. The effects of temperature and strain rate were studied for a sphere moving over a surface using a surface

force apparatus (SFA) and the true friction was found to display time–temperature peaks moving towards high temperature at high strain rate and conversely. In 1959, Bueche and Flom [10] showed that the friction peaks are similar to the loss factor peak in polymers. The influences of the visco-elastic bulk behaviour and interfacial adhesion behaviour were mixed in their experiments. In “rolling ball” (as opposed to “sliding

ball”) experiments, where the interfacial shear stress is negligible, the friction coefficient due to the bulk visco-elasticity also appeared to show peaks similar to those of the bulk loss factor [11]. Recent friction experiments using SFA and AFM techniques have revealed the same type of peaks [12]. At any level, it is quite difficult to separate the influence of the adhesion on the friction from that of the bulk behaviour if a mechanical model of the contact is not available. In a paper comparing SFA and AFM measurements, the effects of the scale of the analysis and size of the sliding tip are discussed [13]. The local pressure in the contact area may vary over three decades and an extremely narrow curvature radius at the interface can lead to an important influence of the surface tension. Hence, in some cases, the normal load may have three origins: applied load, adhesion and surface tension. Depending on the tip size, the friction coefficient may vary over one decade.

1.2. Components of the friction

As mentioned elsewhere [14], the friction coefficient may be split into two components, the adhesion coefficient and the surface deformation coefficient:

$$\mu_{\text{app}} = \mu_{\text{ad}} + \mu_{\text{def}} \quad (1)$$

The contact surface will be symmetric in the case of a perfectly elastic contact, slightly asymmetric for a visco-elastic contact and fully asymmetric for a fully plastic contact. Up until now, no model has described the relationship between the shape of the contact and the material behaviour in the visco-elastic region between perfectly elastic and perfectly plastic behaviour [14]. The effect of strain hardening on the contact area was recently demonstrated by using both of a finite element method [15] and through experiment [16].

One way to separate the friction components is based on the flow line and slip line models developed to describe scratching. Slip lines are assumed to be located where the shear stress has its highest value in the bulk material [17–19]. These models correctly predict three experimentally observed flow processes occurring around a scratching tip: a wave for low contact angles, wear for medium angles and cutting for large angles. However, they make the assumption of a perfectly elastic–plastic material and therefore cannot account for the contact between the rear face of the tip and the surface [20].

Flow lines referred to the motion at the surface of the tip. To separate the components of the friction, they consider the local equilibrium of a moving tip on the surface. The equation of Bowden and Tabor [21] allows calculation of the friction due to the surface deformation in the case of a perfectly elastic–plastic material. In the absence of true friction, for a tip with

an apex angle θ , one has:

$$\mu_{\text{def}} = \frac{2}{\pi} \cot \theta \quad (2)$$

This equation is often employed for low modulus materials like polymers. Using this method, Jardret et al. [22] have developed a flow lines model for a Berkowich tip, taking into account only the flow lines on the front faces of the tip. In the case of low modulus visco-elastic materials, the rear face of the tip remains partly in contact with the surface and the analysis has to be refined. Recently, Bucaille et al. [23] employed an analytical method to take into account the rear face of a conical tip in contact with an elastic–plastic material, which agrees with numerical results obtained with finite element method. Briscoe et al. [24] have developed a flow line model for a perfectly conical tip with a low apex angle, splitting the frontal surface into a stick zone and a slip zone. Subhash and Zhang [25] have developed an analytical vertical flow line model for a conical tip. They showed that the adhesion term and the ploughing term interact in the splitting of the apparent term. Komvopoulos [26] has proposed a flow line model for a conical tip having a spherical extremity and a frontal contact area. The assumptions of the analytical model developed by Tayebi et al. are that the material being ploughed is homogenous with an elastic perfectly plastic behaviour. So the contact area is the front of the spherical tip [27]. Flow lines are inclined and oriented with meridional planes.

In this paper, we present a more refined flow line model based on knowledge of the true contact area between a scratching tip and the surface, which is obtained by “in situ” optical microscopy. Local stress under the contact area is a function of the temperature, the strain rate and the strain. The latter is a function of the friction, the geometry of the tip, the stress–strain relationship, the geometry of the contact and on the external load. These dependencies are the same for the contact pressure and the interfacial shear stress. The contact pressure and the interfacial shear stress may be written as the product of independent functions

$$p = g(T, \dot{\epsilon}) h\left(\frac{a}{R} \text{ or } \theta, \sigma(\epsilon)\right) \quad (3)$$

$$\tau = g(T, \dot{\epsilon}) h\left(\frac{a}{R} \text{ or } \theta, \sigma(\epsilon)\right) f(\mu) \quad (4)$$

where

$$\frac{\tau}{p} = f(\mu) \quad (5)$$

The equations are not model dependent if the shape of the true contact area is exactly known and therefore the apparent friction may be analysed without knowing the stress–strain fields precisely.

2. The model

2.1. True contact area

The contact area between a moving tip and a polymer surface has a front side and a rear side. Fig. 1 shows a photograph of this area obtained with our experimental set-up [29] for a 240 μm radius sphere (estimated after profile recording) moving over a poly-(methacrylate) (PMMA) surface. The shape of the contact area changes with temperature, tip speed and load on the tip. A contact area which is entirely plastically deformed will be called for simplicity a “plastic contact”. If the contact area is not entirely plastically deformed, it will be called an “elastic–plastic contact”.

It was previously shown [30] that the rear and front areas may be predicted in the case of plastic and elastic–plastic contacts on a polymer surface. The rear contact area is due to the elastic recovery of the polymer and depends on the plastic deformation around the contact. Even for a moving tip, the rear area can be almost identical to the front area if the contact is almost elastic and the mechanical loss factor low. The length a_r of the rear contact may be estimated from the length a_f of the front contact and the bulk mechanical properties. In the case of a perfectly elastic contact, the ratio a_r/a_f is 1, while for an elastic–plastic contact, this ratio is [30]:

$$\frac{a_r}{a_f} = 1 - \left[1 - \sqrt{\frac{2cY}{E^*} \tan\theta} \right] \frac{\varepsilon - \varepsilon_e}{\varepsilon - \varepsilon_p} \quad (6)$$

and for a plastic contact [30]:

$$\frac{a_r}{a_f} = \sqrt{\frac{2cY}{E^*} \tan\theta} \quad (7)$$

where Y is the yield stress, E^* the normalised elastic modulus at the contact $\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$, θ the apex

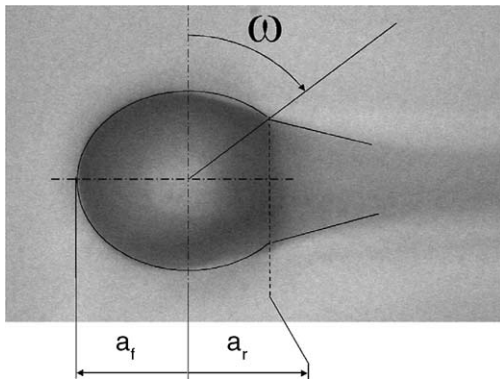


Fig. 1. Photograph of the contact area during scratching. This area has a front part defined by the length a_f and a rear part defined by the angle ω or the length a_r .

angle of the conical scratching tip and c the coefficient connecting the yield stresses measured in a hardness test and under compression ($c \approx 2$ for polymer). ε_e and ε_p denote the contact strains at the end of the elastic contact area and at the beginning of the plastic contact area.

Fig. 2 gives a schematic representation of Eq. (3) as a function of the mean strain ε in the contact area. The strain is proportional to the ratio of the contact radius to the tip radius for a spherical tip, or to the angle of the cone for a conical tip. In the case of a visco-elastic contact, the ratio a_r/a_f is lower than in the elastic case. If the material displays strain hardening, the contact strain decreases and a_r increases. As the strain rate and temperature vary, even under constant loading, the contact area may also vary considerably and the strain near the contact may change from visco-elastic to visco-plastic. The apparent friction coefficient can likewise vary strongly under these circumstances.

Fig. 3 shows the apparent friction for a spherical tip and visco-elastic contact (left), and for a conical tip and visco-plastic contact (right). Fig. 4 illustrates how with a single tip one may obtain a purely visco-elastic contact (g and c), visco-elasto-plastic contact (e and f) or purely visco-plastic contact (a, b and d) simply by changing the load, temperature and tip velocity. The apparent friction can be greater in a sliding process (c) than in a scratching process (d) under certain strain rate and temperature conditions. The groove left on the surface by a visco-elastic sliding relaxes. For very low contact strain, the edges of the visco-elastic groove left on the surface do not remain parallel and the space between them decreases, showing a “flame” shape. The visco-elastic flame relaxes within a time equal to a few

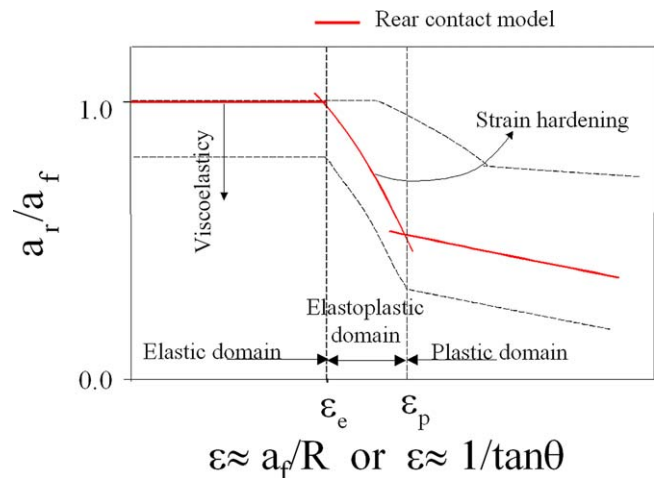


Fig. 2. The ratio of the rear part to the front part of the contact area as a function of the mean strain during contact. This ratio decreases in the case of a visco-elastic body and increases if the material shows strain hardening.

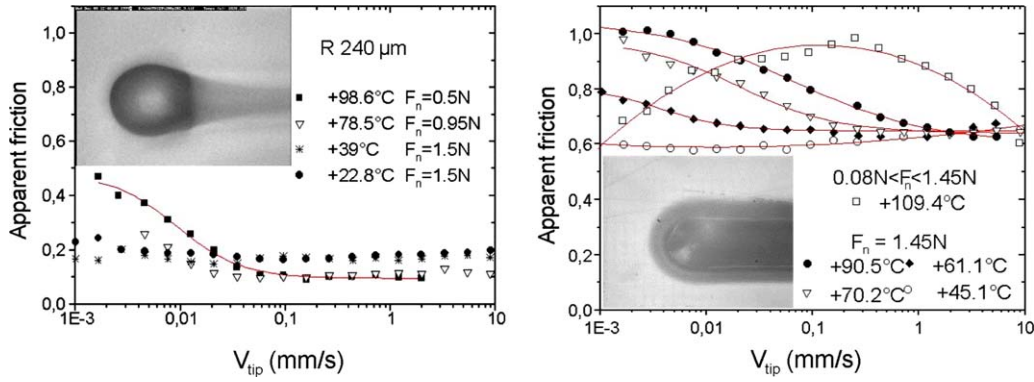


Fig. 3. Apparent friction as a function of the tip velocity, temperature and load. Right: photograph of the contact area during scratching with a cone (120° apex). Left: photograph of the contact area during sliding of a sphere ($240\ \mu\text{m}$ radius). The apparent friction varies by a factor of 10.

times the duration of the contact (c). For a higher contact strain, the groove left on the surface has an imprint of constant width and two lateral slipping faces. Then, the recovery kinetic takes place and the height of the lateral faces as well as the depth decrease (g). During a visco-elasto-plastic contact, the contact strain does not generate a frontal push pad and the lateral plastic pads generated by the visco-elastic unloading appear at the edge of the groove. For a visco-plastic contact, the polymer flows visco-plastically around the indenter and the frontal and lateral pads join to form a continuous cord.

2.2. Two-dimensional model

In the ideal case of two perfectly flat and smooth planes sliding without local irreversible deformation,

the apparent friction μ_{app} is identical to the true local friction μ . On real rough surfaces, when material on one surface flows around the contact points, the apparent friction is the sum of the true local friction and the geometrical ploughing friction. The way these two parameters interact is easily demonstrated in a two-dimensional model as shown in Fig. 5.

In the most general case, the contact surface is smaller at the rear face than at the front face of the tip. It is assumed that the contact pressure at the two faces is the same and that the scission is linear with pressure. Using θ for the half cone angle and $\chi = b/a$ to characterise the symmetry of the contact surface, the force equilibrium leads to the following equations:

$$F_n = p(S_f + S_d)\sin\theta + \tau(S_d - S_f)\cos\theta \quad (8)$$

$$F_t = p(S_f - S_d)\cos\theta + \tau(S_d + S_f)\sin\theta \quad (9)$$

It was shown that the interfacial shear stress depends on the contact pressure, the temperature, the contact

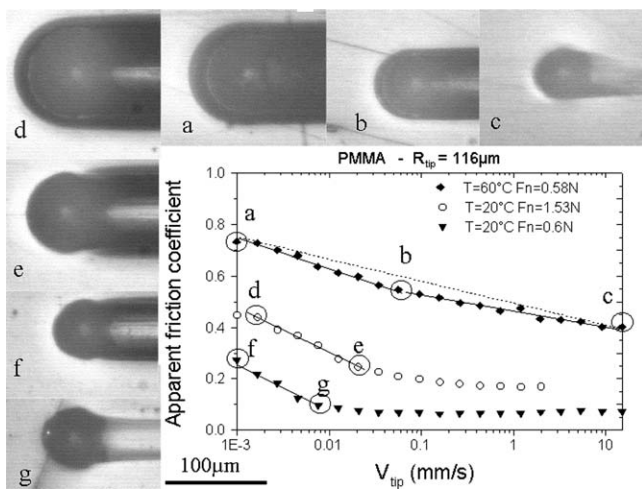


Fig. 4. Photographs of the contact area under experimental conditions ranging from scratching to sliding using a spherical tip of $116\ \mu\text{m}$ radius. The shape of the contact area and the apparent friction are not simply correlated (a and d). A visco-elastic contact area (c) may display higher friction than a visco-plastic (d) or a visco-elasto-plastic contact area (e and f).

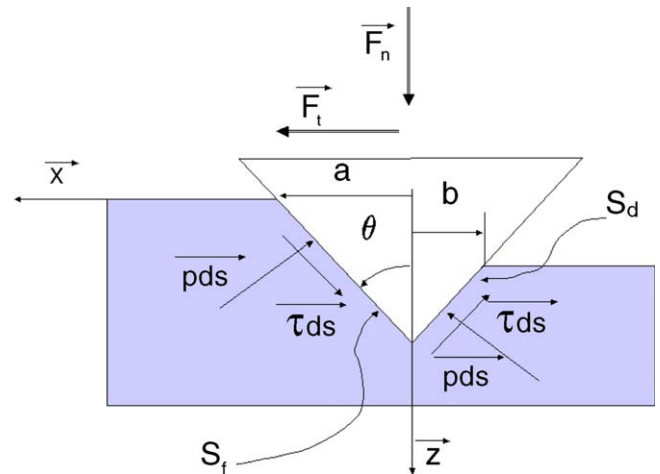


Fig. 5. Stresses acting on a moving two-dimensional tip. $\chi = b/a$ describes the symmetry of the contact area.

time and the interfacial strain rate [28]. An empirical relationship is

$$\tau = \tau_0 + \alpha p \quad (10)$$

where p is the contact pressure, α and τ_0 are constant for a given polymer when all other contact variables are fixed. With this assumption, the local friction may be defined as:

$$\mu = \frac{\tau_0}{p} + \alpha \quad (11)$$

Briscoe [28] gives values for α and τ_0 for many solid polymers: for example, for PMMA $\alpha = 0.36$ and $\tau_0 = 10$ MPa, while scratch hardness is typically equal to 250 MPa at room temperature. So at first approximation, $\mu \approx \alpha(t, T)$ and this approximation remains true for many solid glassy polymers.

Substituting $S_d = \chi S_f$ and $\tau = \mu p$, the apparent friction yields (Fig. 6):

$$\frac{F_t}{F_n} = \frac{(1 - \chi) + \mu(1 + \chi)\tan\theta}{(1 + \chi)\tan\theta + \mu(\chi - 1)} \quad (12)$$

There are some specific cases, which are of particular interest. Thus, for a perfectly elastic contact, $\chi = 1$ and $F_t/F_n = \mu$, while for a perfectly plastic contact, $\chi = 0$ and $F_t/F_n = (1 + \mu\tan\theta)/(\tan\theta - \mu)$. If there is no true friction, $F_t/F_n = 1/\tan\theta$ is the geometrical friction. Finally, for the particular value of $\mu = \tan\theta(1 + \chi)/(1 - \chi)$, the apparent friction tends to infinity (Eq. (12)). Fig. 7 shows the values of χ at which the apparent friction becomes infinite. Since for most materials μ is less than 1, the apparent friction may become very large only for low values of χ and θ .

2.3. Three-dimensional model: three types of flow lines

The contact area is divided into a front and rear part and the latter may be defined by an angle ω obtained from $\tan\omega = a_r/l$, where l is the half width of the groove behind the tip as shown in Fig. 1. This rear part of the contact area must be taken into account to calculate the true friction coefficient. Models of the flow lines along the front and rear parts of the contact area

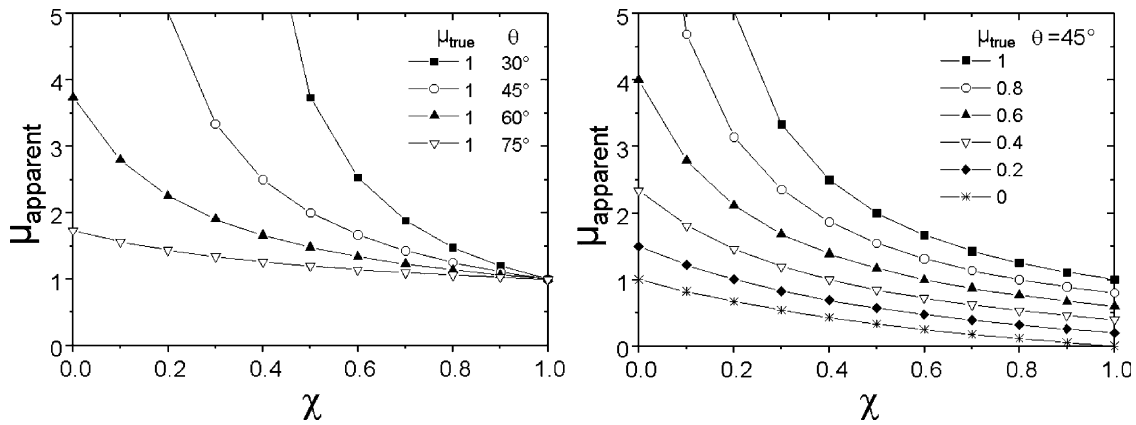


Fig. 6. Apparent friction as a function of the contact symmetry χ for several cone angles and true friction coefficients.

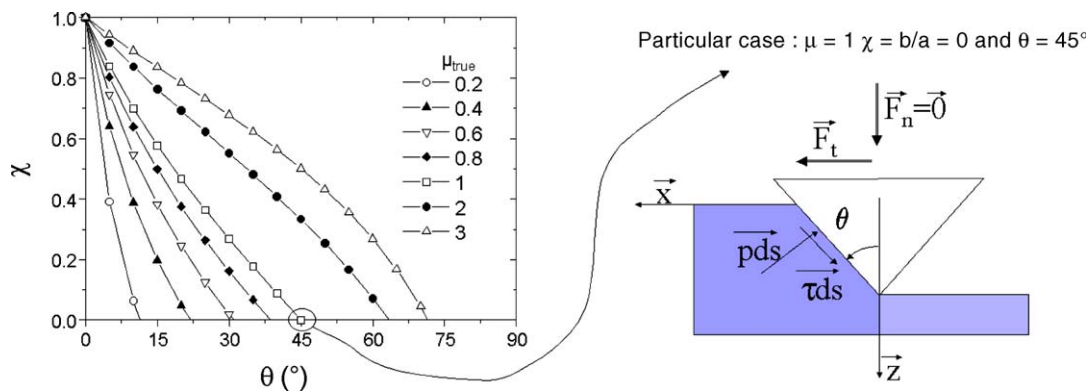


Fig. 7. Contact symmetry χ as a function of the geometry of the tip, for several cone angles, given $\vec{F}_n = \vec{0}$.

must satisfy several conditions: for a symmetrical contact, the friction due to geometrical effects must vanish, the lines should not exhibit discontinuities and the extension of the model of Tabor should be applicable at any value of ω . Three simplified models will be examined here: horizontal lines, vertical lines and an intermediate case.

Fig. 8 depicts the three types of flow lines model tested. The horizontal lines are the intercept between a horizontal plane and the tip and correspond to circles for a uniaxial tip. Vertical lines are the intercept between a vertical plane and the tip and correspond to hyperboles in the case of a conical tip. The intermediate case, called “secant lines”, is obtained from the intercept of the tip with a plane perpendicular to the lateral generatrix. These lines are elliptical for cones and circles for spheres.

The local normal and tangential forces due to material flow acting on the tip are:

$$\vec{N} = -pds\vec{n} \quad (13)$$

$$\vec{T} = \tau ds\vec{t} \quad (14)$$

where $\vec{n}$ and $\vec{t}$ are unit normal and tangent vectors and p and τ the local normal pressure and scission stress. The mean macroscopic values of the forces may be defined by:

$$\vec{F}_n = F_n\vec{z} = [(Ap + B\tau)S_n]\vec{z} \quad (15)$$

$$\vec{F}_t = F_t\vec{x} = [(Cp + D\tau)S_n]\vec{x} \quad (16)$$

with

$$\begin{aligned} A &= \frac{1}{S_n} \int \vec{n} \cdot \vec{z} \, ds & B &= -\frac{1}{S_n} \int \vec{t} \cdot \vec{z} \, ds \\ C &= \frac{1}{S_n} \int \vec{n} \cdot \vec{x} \, ds & D &= \frac{1}{S_n} \int \vec{t} \cdot \vec{x} \, ds \end{aligned} \quad (17)$$

and S_n is the normal projected contact area. The true friction μ is defined as $\mu = \tau/p$ and therefore:

$$\frac{F_t}{F_n} = \mu_{app} = \frac{C + D\mu}{A + B\mu} \quad (18)$$

A , B , C and D do not depend on the size of the contact area, but depend on the shape of the tip and the shape of the contact area. The latter is dependent on the stress–strain law of the material as previously discussed at the end of the introduction. Details of the calculation are given in Appendix A.

The local stress $\vec{N} = -pds\vec{n}$ depends only on the shape of the tip and the local pressure. In a first step, the local pressure was assumed to be constant $p = p_0$ over the whole contact area. Recent numerical simulation showed that the stress along the frontal contact line is similar to the stress along the lateral generatrix [25]. Local pressure was considered to be non-uniform

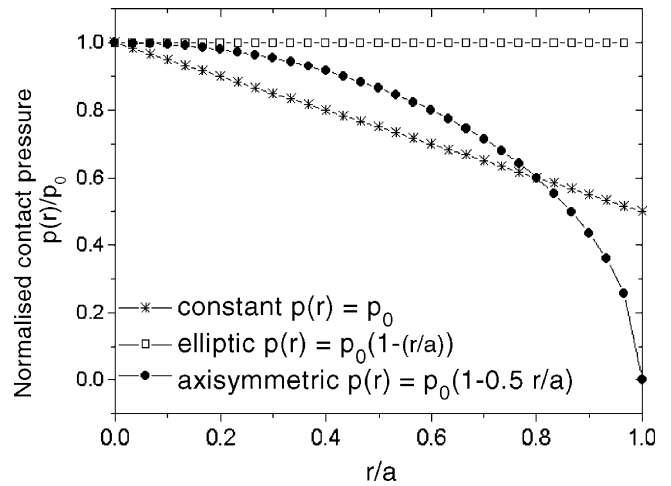


Fig. 9. Three cases of the local pressure acting in the contact area: constant, elliptical and with uniaxial symmetry ($\varphi = 0.5$).

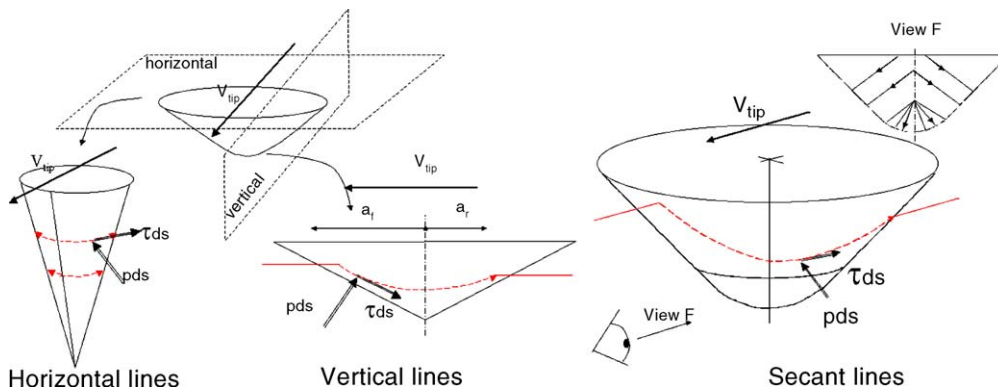


Fig. 8. Sketch of the flow lines of the models: horizontal lines (left), vertical lines (centre) and intermediate case (right).

and elliptical $p(r) = p_0(1 - (r/a)^2)^{1/2}$ or uniaxial $p(r) = p_0(1 - \varphi(r/a))$ where $0 < \varphi < 1$. Elastic and viscoelastic contacts were simulated [31] as likewise perfectly plastic contacts [32] and results are presented in Fig. 9.

3. Results and discussion

3.1. Simplified material: true local friction is zero

If the true friction is zero, the results do not depend on the flow line model and it is possible to demonstrate some general properties, which are not flow lines dependent. The apparent friction is then equal to the geometrical friction $B = D = 0$, while C/A is the Tabor friction coefficient [21] $\mu_{\text{def}} = 2/\pi \cot \theta$. In the case of a conical tip with a spherical extremity, the simulated friction may be plotted as a function of the contact radius, normalised by the radius at the junction between the sphere and the cone $R_c = R \cos \theta$, where θ

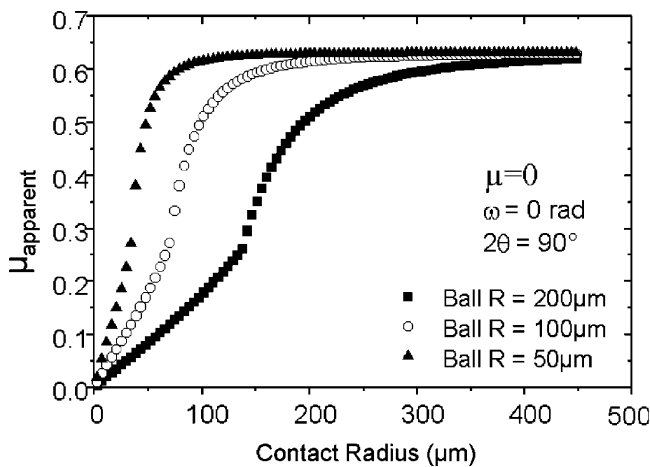


Fig. 10. Apparent friction versus the contact radius for several radii of the sphere. The cone angle is 90° and the true local friction and rear contact area (angle ω) are zero.

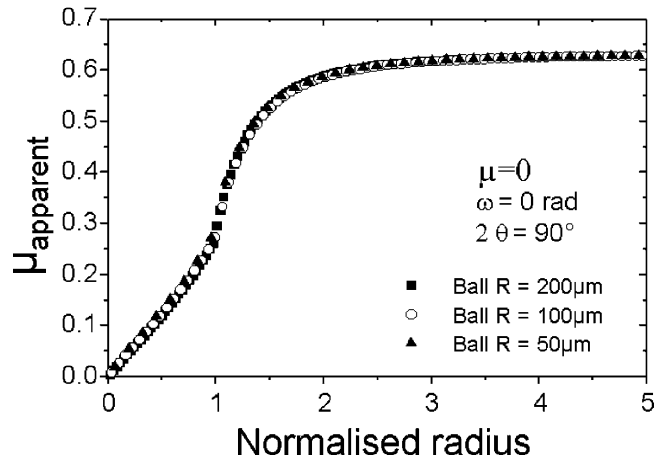


Fig. 11. Apparent friction versus the normalised contact radius for several radii of the sphere. The cone angle is 90° and the true local friction and rear contact area (angle ω) are zero.

is the half angle of the cone:

$$R_{\text{norm}} = \frac{a}{R_c} \quad (19)$$

Figs. 10 and 11, which show simulations with a true friction equal to zero, clearly demonstrate that the results do not depend on the radius of the sphere when plotted as a function of the normalised radius. Simulations for several cone angles performed using the same assumptions are shown in Fig. 12.

3.2. Realistic perfectly plastic material

Simulations of the apparent friction in a perfectly plastic material ($\omega = 0$), for the three flow line models and several cone angles and true friction coefficients, are presented in Figs. 13–20. The general trends of the three models can be inferred from these figures.

3.2.1. Horizontal and vertical flow line models

The model based on vertical flow lines diverges for low cone angles and the apparent friction is largely

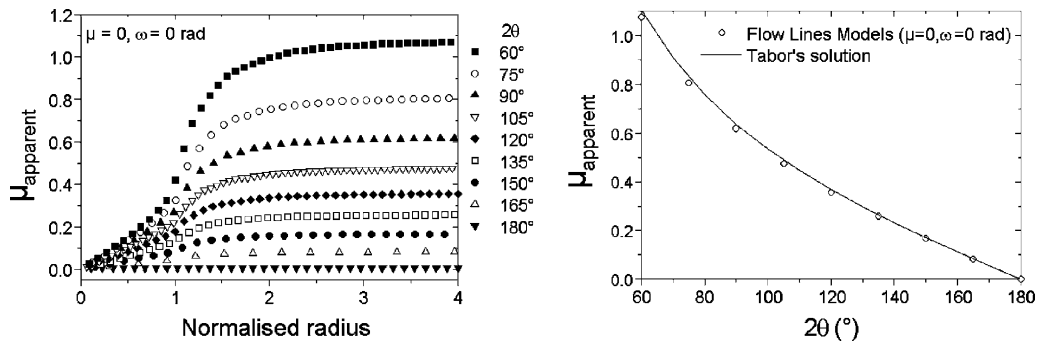


Fig. 12. Apparent friction for several cone angles. The true local friction and rear contact area (angle ω) are zero and asymptotic values are consistent with the results of Tabor.

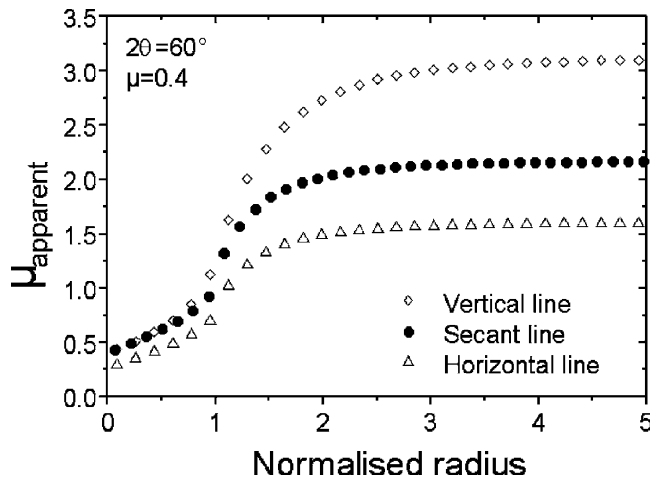


Fig. 13. Apparent friction for a cone angle of 60° and a true friction coefficient of 0.4.

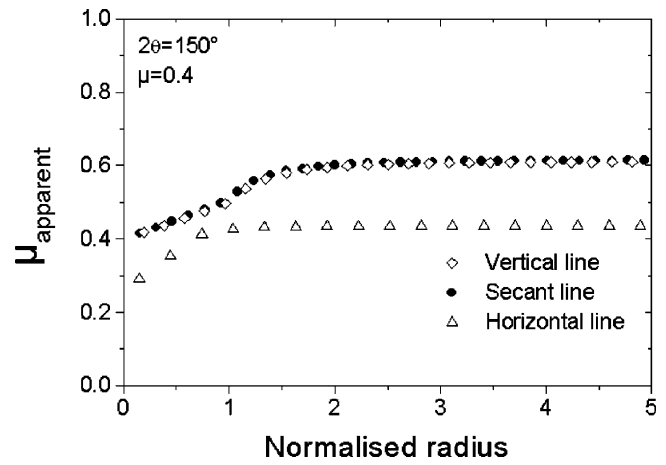


Fig. 16. Apparent friction for a cone angle of 150° and a true friction coefficient of 0.4.

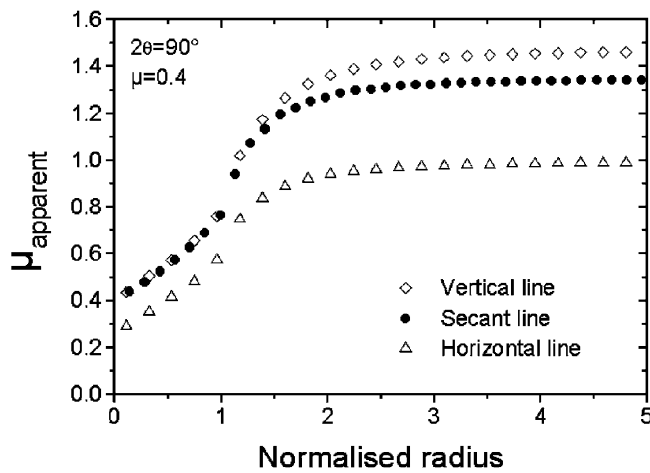


Fig. 14. Apparent friction for a cone angle of 90° and a true friction coefficient of 0.4.

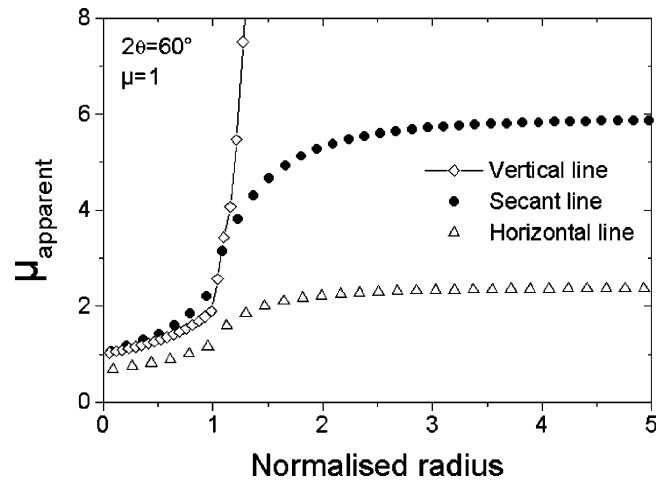


Fig. 17. Apparent friction for a cone angle of 60° and a true friction coefficient of 1.

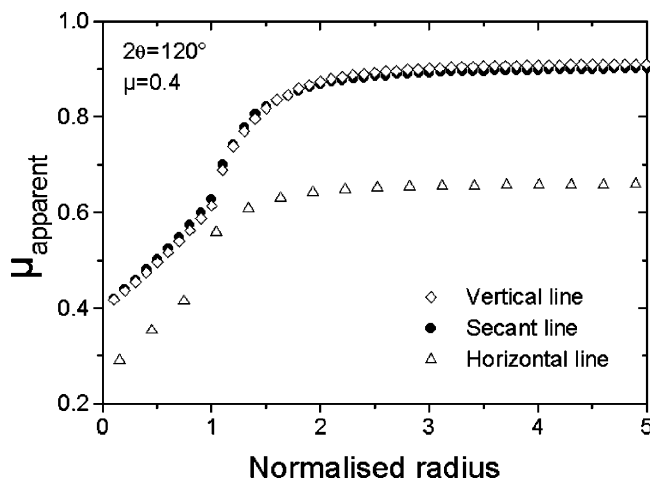


Fig. 15. Apparent friction for a cone angle of 120° and a true friction coefficient of 0.4.

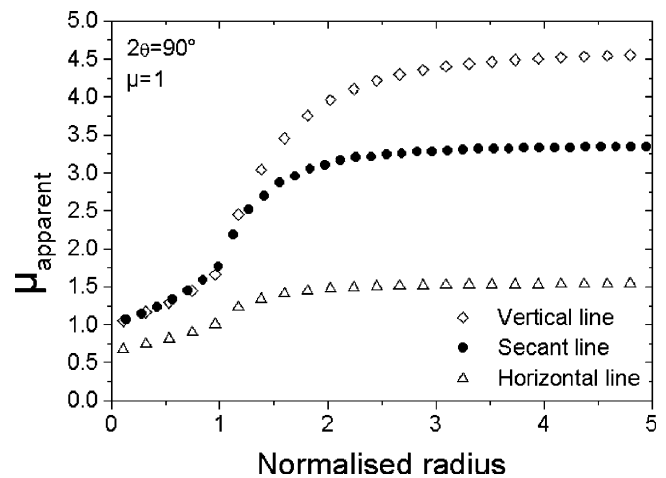


Fig. 18. Apparent friction for a cone angle of 90° and a true friction coefficient of 1.

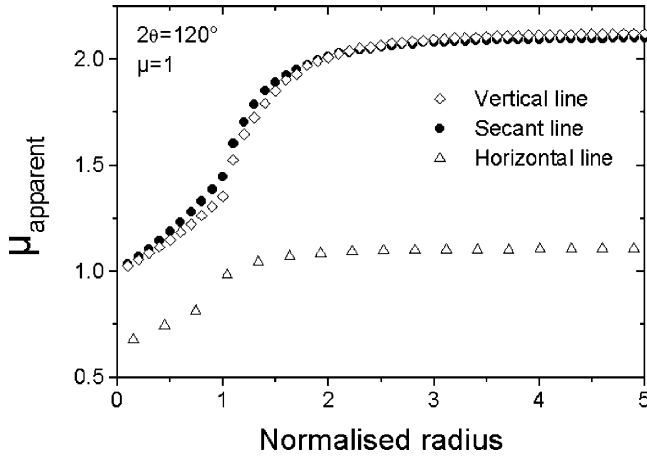


Fig. 19. Apparent friction for a cone angle of 120° and a true friction coefficient of 1.

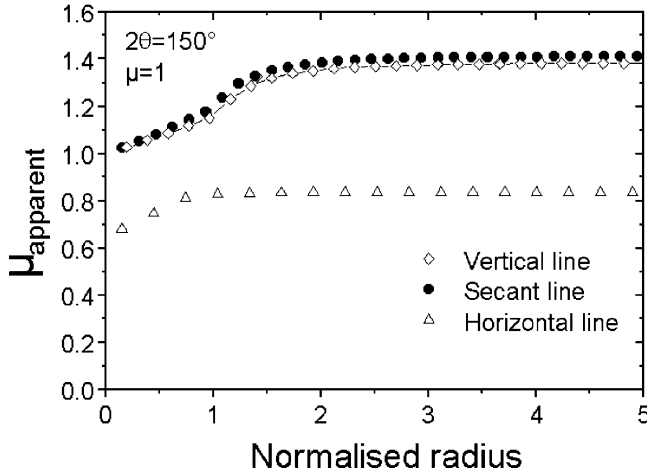


Fig. 20. Apparent friction for a cone angle of 150° and a true friction coefficient of 1.

overestimated with respect to Bowden and Tabor calculations. The horizontal lines model, which is commonly used in many analytical approaches, underestimates the friction for low values of the normalised contact radius and the apparent friction would seem to be lower than the true friction. This model was initially developed to estimate the friction of a sharp tip (very low cone angle) and is not suitable for spheres having a rather large “apparent cone angle”. Fig. 21 shows the situation when the contact radius tends to zero. The apparent friction may then be estimated as:

$$F_n = S_n p = \frac{\pi a^2 p}{2} \quad (20)$$

$$F_t = \int \tau \cos \delta \, ds = a^2 \tau \quad (21)$$

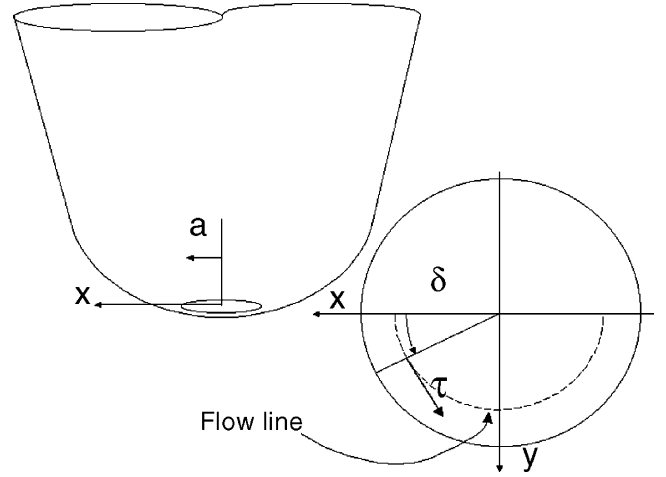


Fig. 21. Local scission and horizontal flow lines as the contact radius tends to zero.

and hence

$$\mu_{app} = \frac{F_t}{F_n} = \frac{2}{\pi} \frac{\tau}{p} \quad (22)$$

Values calculated from Eq. (22) are close to those obtained using the horizontal flow line model.

3.2.2. Secant flow lines model

The third flow line model described in Fig. 8 tends to the case of horizontal lines for very low cone angles and to that of vertical lines for large cone angles. It likewise approaches vertical lines for low values of the normalised contact radius and/or large cone angles if the true friction is low. Furthermore, it appears in Fig. 22 that a fraction of the local scission will contribute to the normal force $\bar{F}_n$.

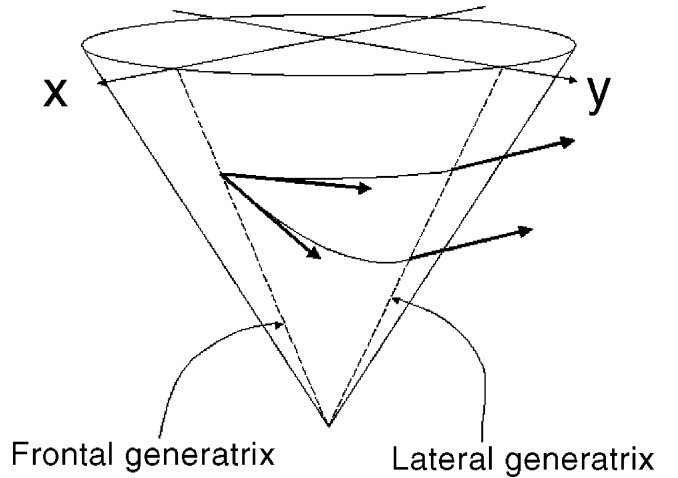


Fig. 22. Local scission for the intermediate flow lines model. Part of the local interfacial shear stress acts on the normal force $\bar{F}_n$ and part of the interfacial shear stress has a lateral component.

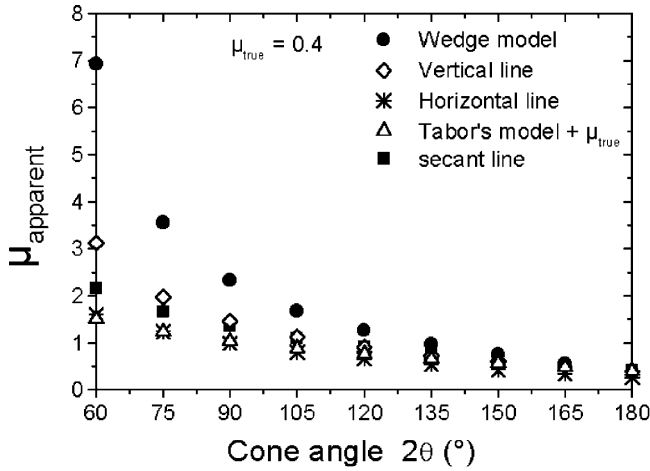


Fig. 23. Limiting apparent friction predicted by different flow lines models for $a/R_c = \infty$, $\omega = 0$, and a true local friction of 0.4.

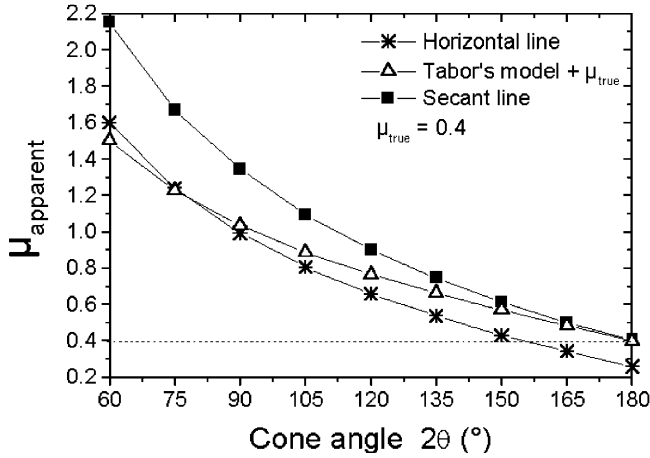


Fig. 24. Zoom of the limiting apparent friction predicted by different flow lines models versus the cone angle.

The limiting apparent friction for $a/R_c = \infty$ and a true local friction of 0.4 is plotted in Figs. 23 and 24. Simulations were performed for several cone angles using the three flow line models and also the usual estimation of the apparent friction (Tabor model + true friction):

$$\mu_{app} = \mu + \mu_{def} \quad (23)$$

where μ_{def} is given by Eq. (2) and the two-dimensional model (Eq. (12)) where $\chi = 0$:

$$\mu_{app} = \frac{\cos\theta + \mu\sin\theta}{\sin\theta - \mu\cos\theta} \quad (24)$$

and the true local friction coefficient is $\mu = \tau/p$. Secant flow lines clearly provide the best approximation.

Fig. 25 shows the apparent friction in a purely plastic material ($\omega = 0$) for various true friction coefficients

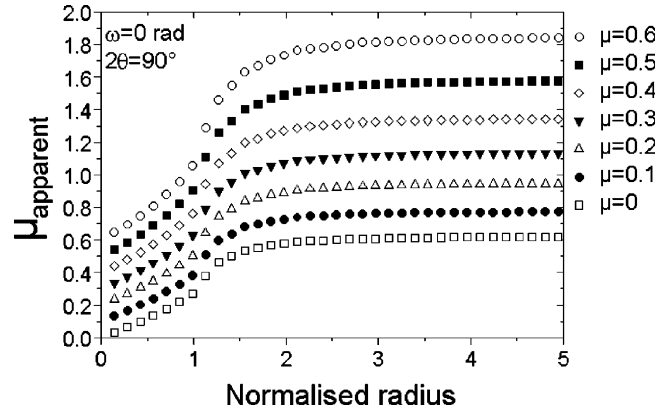


Fig. 25. Apparent friction in a purely plastic material for several true friction coefficients, $\omega = 0$ and a cone angle of 90° .

and a cone angle of 90° . The apparent friction is in all cases equal to the true local friction at $a/R_c = 0$ (very low applied normal force) and tends to a limiting value for $a/R_c > 3$ (high applied normal force). The increase in apparent friction with applied load is roughly the same for all true local friction coefficients.

3.3. Realistic elastic–plastic material

If the material is partly elastic, the contact area has not only a front part but also a rear part ($\omega > 0$). Fig. 26 shows how the rear part of the contact area influences the apparent friction coefficient. When $\omega = \pi/2$ (symmetrical contact), the apparent and true frictions are the same. The apparent friction at $a/R_c = \infty$ is plotted in Fig. 27 as a function of the radius r and rear contact angle ω , for the three models of the pressure plotted in Fig. 9 (constant, elliptical or uniaxial) acting on the contact surface.

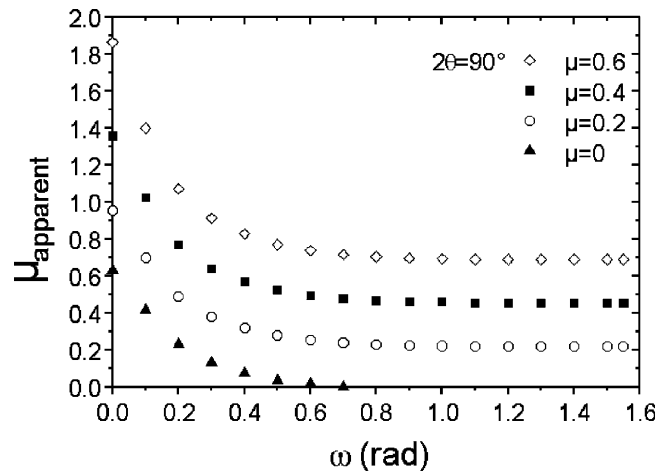


Fig. 26. Apparent friction in an elastic–plastic material for several true friction coefficients, $a/R_c = \infty$, and a cone angle of 90° .

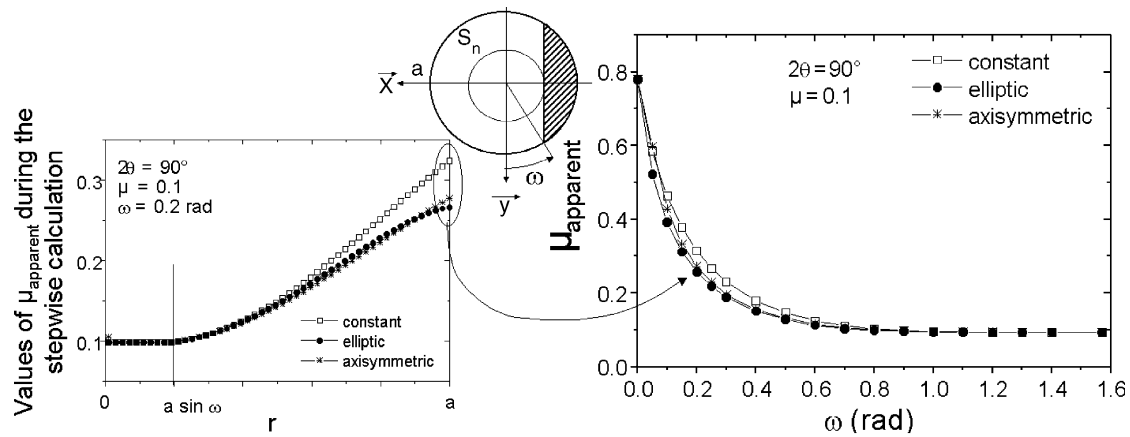


Fig. 27. Apparent friction at $a/R_c = \infty$ as a function of the rear contact angle ω , for the three models of the pressure acting on the contact surface ($\varphi = 0.5$ for the axisymmetric contact pressure).

There is very little difference between the three results. In fact, two situations may be considered, firstly where there is no true local friction and the apparent friction does not depend on the type of pressure for a symmetrical frontal contact. Secondly, if the rear and front parts of the contact are the same, the apparent friction is equal to the true friction and the type of pressure again has no influence. As shown in Fig. 27, only a small part of the contact surface (shaded) is not covered by these two situations and therefore the type of pressure acting in the contact area has little influence on the apparent friction. Conversely, if the apparent friction coefficient and shape of the tip are known from experimental data, the true local friction may be calculated from Eq. (15) thus:

$$\mu = \frac{A\mu_{app} - C}{D - B\mu_{app}} \quad (25)$$

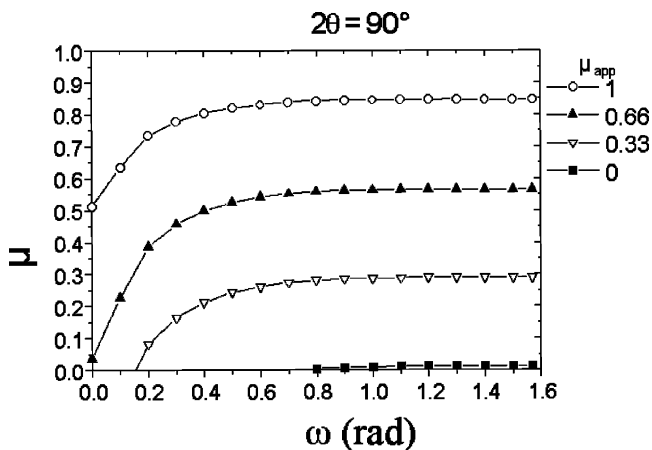


Fig. 28. True local friction coefficient calculated from the apparent friction as a function of the rear contact area. The cone angle is 90° and $a/R_c = \infty$.

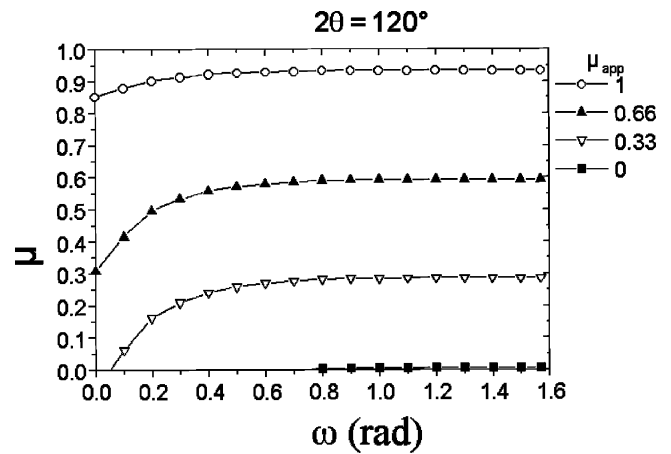


Fig. 29. True local friction coefficient calculated from the apparent friction as a function of the rear contact area. The cone angle is 120° and $a/R_c = \infty$.

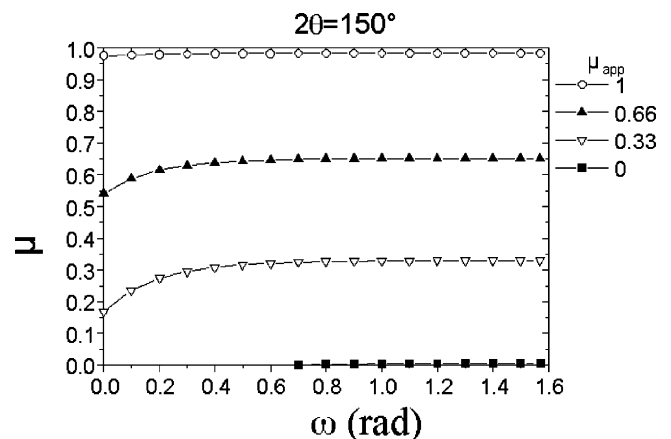


Fig. 30. True local friction coefficient calculated from the apparent friction as a function of the rear contact area. The cone angle is 150° and $a/R_c = \infty$.

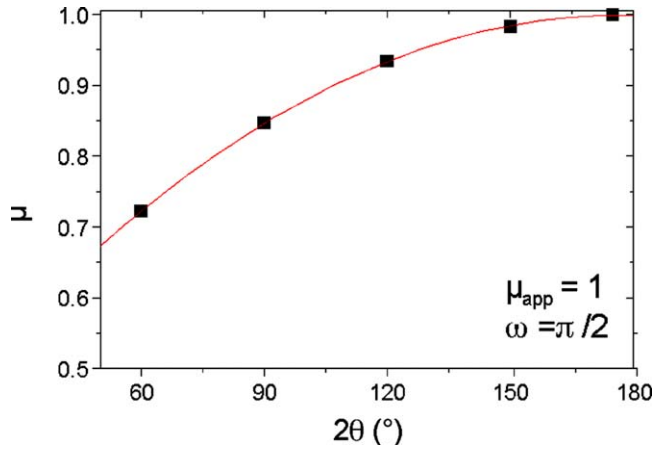


Fig. 31. True local friction coefficient calculated as a function of the apex angle of a conical tip. The apparent friction is 1, $\omega = \pi/2$ and $a/R_c = \infty$.

Figs. 28–31 demonstrate how this may be done and also reveal an intrinsic error of the model. Thus, for $\omega = \pi/2$, it is acknowledged that the apparent and true frictions should be the same. The discrepancy is nevertheless fairly small and remains below 5% provided the apex angle of the tip is greater than 120° , which is in agreement with experimental observations of the sliding of a conical or ball tip on a polymeric material. If the apex angle is less than 90° , the discrepancy may exceed 15% (Fig. 31) and should be attributed to the component t_y , directly linked to this angle and given in Appendix A.2.3. t_y is the lateral component of the interfacial shear stress schematically shown in Fig. 22. If the tip is symmetrical and the material homogenous, the lateral load is equal to zero.

4. Conclusion

Three types of flow lines were tested in a new analytical simulation model designed to determine the apparent friction coefficient of conical and spherical tips scratching a surface. The input data required by the simulation are the true contact area, the true local friction coefficient and a model of the pressure acting on the contact surface. The type of pressure introduced into the model does not significantly modify the results of the simulation and the secant flow line model seems to be a good model. Conversely, when the apparent friction is known from experimental data, it is possible to deduce the true local friction coefficient at the interface between a tip and the material.

Appendix A. Calculation of $\vec{n}$ and $\vec{t}$

A.1. The normal unit vector $\vec{n}$, Fig. A.1

The components of $\vec{n}$ are independent of the flow lines.

On the cone,

$$\vec{n} = \cos\theta\cos\beta\vec{x} + \cos\theta\sin\beta\vec{y} + \sin\theta\vec{z}$$

On the spherical tip,

$$\vec{n} = \sin\alpha\vec{x} - \cos\alpha\sin\gamma\vec{y} + \cos\alpha\cos\gamma\vec{z}$$

and

$$\cos^2\theta = \sin^2\alpha + \cos^2\alpha\sin^2\gamma$$

A.2. The tangential unit vector $\vec{t}$, Fig. A.2

The components of $\vec{t}$ are dependent on the flow lines.

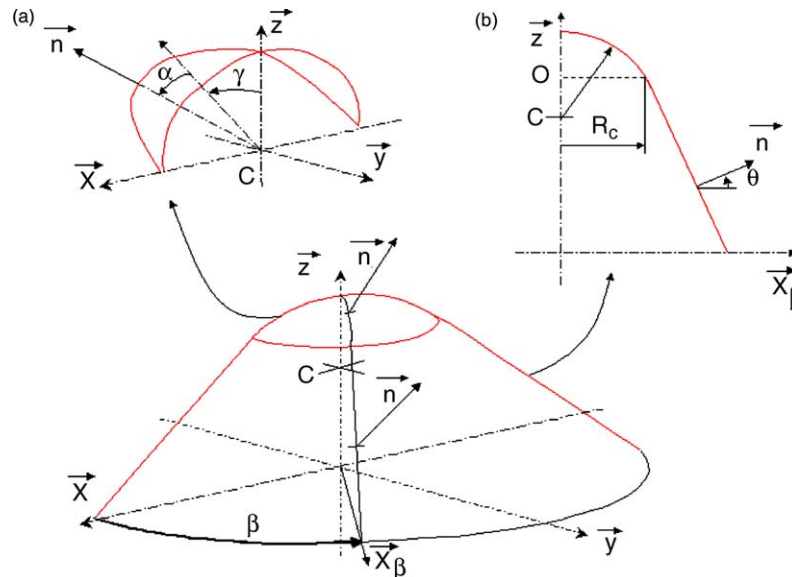


Fig. A.1. Detailed geometry of the tip: (a) the spherical extremity, (b) the cone.

A.2.1. Horizontal flow lines

For the spherical tip and the cone,

$$\vec{t} = -\sin\beta\vec{x} + \cos\beta\vec{y}$$

A.2.2. Vertical flow lines

The conditions are $(\vec{t} \cdot \vec{n} = 0)$ and $t_y = 0$.

For the cone tip,

$$t_z \sin\theta + t_x \cos\theta \cos\beta = 0$$

$$\vec{T} = \begin{pmatrix} -1 \\ 0 \\ \cos\beta \\ \tan\theta \end{pmatrix} \quad \vec{t} = \begin{pmatrix} -\tan\theta \\ (\tan^2\theta + \cos^2\beta)^{0.5} \\ 0 \\ \cos\beta \\ (\tan^2\theta + \cos^2\beta)^{0.5} \end{pmatrix}$$

For the spherical tip,

$$t_x \sin\alpha + t_z \cos\alpha \cos\gamma = 0$$

$$\vec{T} = \begin{pmatrix} -1 \\ 0 \\ \tan\alpha \\ \cos\gamma \end{pmatrix} \quad \vec{t} = \begin{pmatrix} -\cos\gamma \\ (\tan^2\alpha + \cos^2\gamma)^{0.5} \\ 0 \\ \tan\alpha \\ (\tan^2\alpha + \cos^2\gamma)^{0.5} \end{pmatrix}$$

A.2.3. Secant flow lines

For the spherical tip,

$$\vec{t} = \begin{pmatrix} -\cos\alpha \\ -\sin\alpha\sin\gamma \\ +\sin\alpha\cos\gamma \end{pmatrix}$$

For the cone tip, $\vec{t} \cdot \vec{n} = 0$ and $\vec{t}$ lies in the plane perpendicular to the lateral generatrix.

$$\vec{t} = \begin{pmatrix} -\cos\alpha_t \\ -\sin\alpha_t \cos\theta \\ +\sin\alpha_t \sin\theta \end{pmatrix}$$

$$\tan\alpha_t = \frac{\cos\theta \cos\beta}{\cos^2\theta \sin\beta + \sin^2\theta}$$

A.3. Integrals

A.3.1. On the spherical tip

$$\gamma < \gamma_1$$

$$\tan\gamma_1 = \frac{R_c}{\sqrt{R^2 - R_c^2}}$$

$$\alpha < \alpha_1$$

$$R_c^2 = R^2(\sin^2\alpha_1 + \cos^2\alpha_1 \sin^2\gamma)$$

$$\tan^2\alpha_1 = \frac{R_c^2}{R^2 \cos^2\alpha_1} - \sin^2\gamma$$

A.3.2. On the cone

$$\beta < \omega.$$

A.3.3. On the rear part of the contact area

Below $a \sin \omega$ (inner circle in Fig. 27), the contact area for the integral calculation is a circle. Above $a \sin \omega$, $a_r = -2\alpha_1\omega/\pi$. On the cone, $\beta < \omega + \pi/2$.

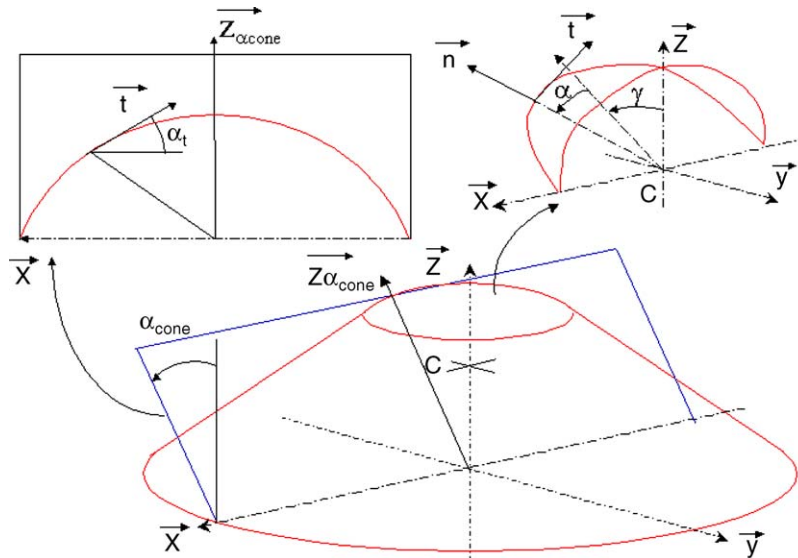


Fig. A.2. Flow lines at the spherical tip and the cone ($\alpha_{\text{cone}} = \pi/2 - \theta$).

References

- [1] Bowden FP, Tabor D. The friction and lubrication of solids. Oxford: Clarendon Press; 1954.
- [2] Greenwood JA, Williamson JBP. Contact of nominally flat surfaces. *Proc Roy Soc A* 1966;295:300–19.
- [3] Johnson KL, Kendall K, Roberts AD. Surface energy and the contact of elastic solids. *Proc Roy Soc A* 1971;324:301–13.
- [4] Dejarguin BV, Muller VM, Toporov YuP. Effect of contact deformations on adhesion of particles. *J Colloid Interface Sci* 1975;53:314–26.
- [5] Fuller KNG, Tabor D. Effect of surface-roughness on adhesion of elastic solids. *Proc Roy Soc A* 1975;345:327–42.
- [6] Maugis D. On the contact and adhesion of rough surfaces. *J Adhes Sci Technol* 1996;10:161–75.
- [7] Baumberger T, Berthoud P, Caroli C. Physical analysis of the state- and rate-dependant friction law II. Dynamic friction. *Phys Rev B* 1999;60:3928–39.
- [8] Chang WR, Etsion I, Bogy DB. Static friction coefficient model for metallic surfaces. *ASME J Tribol* 1988;110:57–63.
- [9] Liu Z, Sun J, Shen W. Study of plowing and friction at the surfaces of plastic deformed metals. *Tribol Int* 2002;35:511–22.
- [10] Bueche AM, Flom DG. Surface friction and dynamic mechanical properties of polymers. *Wear* 1959;2:168–82.
- [11] Ludema KC, Tabor D. The friction and visco-elastic properties of polymeric solids. *Wear* 1966;9:329–48.
- [12] Yoshizawa H, Chen Y-L, Israelachvili J. Fundamental mechanisms of interfacial friction. 1. Relation between adhesion and friction. *J Phys Chem* 1993;97:4128–40.
- [13] McGuiggan PM, Zhang J, Hsu SM. Comparison of friction measurements using the atomic force microscope and the surface forces apparatus: the issue of scale. *Tribol Lett* 2001;10:217–23.
- [14] Moore DF. The friction and lubrication of elastomers. Oxford: Pergamon Press; 1972.
- [15] Bucaille J-L. PhD Ecole nationale supérieur des mines de Paris; 2001.
- [16] Bucaille J-L, Gauthier C, Felder E, Schirrer R. The influence of strain hardening of polymers on the push pad formation in scratch tests: experiments and numerical modelling. *J Polym Sci B*, submitted for publication.
- [17] Childs THC. The sliding of rigid cones over metals in high adhesion conditions. *Int J Mech Sci* 1970;12:393–403.
- [18] Challen JM, Oxley PLB. An explanation of the different regimes of friction and wear using asperity deformation models. *Wear* 1979;53:229–43.
- [19] De Vathaire M, Delamare F, Felder E. An upper bound model of ploughing by a pyramidal indenter. *Wear* 1981;66:55–64.
- [20] Lovell MR, Deng Z. Experimental investigation of sliding friction between hard and deformable surfaces with application to manufacturing processes. *Wear* 1999;236:117.
- [21] Bowden FP, Tabor D. Friction, lubrication and wear: a survey of work during the last decade. *Brit J Appl Phys* 1966;17: 1521–44.
- [22] Jardret V, Zahouani H, Loubet J-L, Mathia TG. Understanding and quantification of elastic and plastic deformation during a scratch test. *Wear* 1998;218:8–14.
- [23] Bucaille J-L, Felder E, Hochstetter G. Mechanical analysis of the scratch test on elastic and perfectly plastic materials with the three-dimensional finite element modelling. *Wear* 2001;249:422–32.
- [24] Briscoe BJ, Biswas SK, Sinha SK, Panesar SS. The scratch hardness and friction of a soft rigid-plastic solid. *Tribol Int* 1993;26:183–93.
- [25] Subhash G, Zhang W. Investigation of the overall friction coefficient in single-pass scratch test. *Wear* 2002;252:123–34.
- [26] Komvopoulos K. Sliding friction mechanisms of boundary-lubricated layered surfaces. 2. Theoretical-analysis. *Tribol Trans* 1991;34:281–91.
- [27] Tayebi N, Conry TF, Polycarboou AA. Determination of hardness from nanoscratch experiments: correction for interfacial shear stress and elastic recovery. *J Mater Res* 2003;18: 2150–62.
- [28] Briscoe BJ. Interfacial friction of polymer composites. In: Friedrich K, editor. General fundamental principles. Composite materials series, vol. 1. Elsevier; 1986 (Chapter 2).
- [29] Gauthier C, Schirrer R. Time and temperature dependence of the scratch properties of poly(methylmethacrylate) surfaces. *J Mater Sci* 2000;35:2121–30.
- [30] Gauthier C, Lafaye S, Schirrer R. Elastic recovery of a scratch in a polymeric surface: experiments and analysis. *Tribol Int* 2001;34:469–79.
- [31] Johnson KL. Contact mechanics. Cambridge University Press; 1985.
- [32] Chitkara NR, Butt MA. Numerical construction of axisymmetrical slip-line fields for indentation of thick blocks by rigid conical indenters and friction at the tool metal interface. *Int J Mech Sci* 1992;34:849–62.